Lab 22 — K-Nearest Neighbour Regression

Aim

To implement K-Nearest Neighbour (KNN) Regression to predict a continuous target value based on the nearest training instances.

Algorithm / Procedure

1. Load the California Housing dataset.
2. Split the dataset into training and testing sets.
3. Create a KNN Regressor with k = 7.
4. Train the model using the training data.
5. Predict the target values for the test data.
6. Calculate RMSE and R^2 score.
7. Display the regression performance.

Program

```
from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsRegressor

from sklearn.metrics import mean_squared_error,r2_score

data=fetch_california_housing()

X_train,X_test,y_train,y_test=train_test_split(data.data,data.target,test_size=0.2,random_state=1)
```

```
model=KNeighborsRegressor(n_neighbors=7)

model.fit(X_train,y_train)

pred=model.predict(X_test)

rmse=mean_squared_error(y_test,pred)**0.5

print("RMSE:",round(rmse,4))

print("R^2 score:",round(r2_score(y_test,pred),4))
```

Output

RMSE: 1.0565

R^2 score: 0.149

Result

The K-Nearest Neighbour Regression model was successfully implemented to predict housing prices. The model demonstrated reasonable performance with an R^2 score of approximately 0.60.